FEDERAL SPECIFICATION

ISOBUTYL ACETATE; (FOR USE IN ORGANIC COATINGS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the requirements for isobutyl acetate for use in lacquer thinners.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards.—The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Standard No. 141—Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publications form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Society for Testing Materials Publication:

ASTM Standards — Part 4.

(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

Uniform Freight Classification Ratings, Rules, and Regulations.

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(Copies may be obtained from the Uniform Freight Classification Committee, 1 Park Avenue, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Quantitative requirements.—The isobutyl acetate shall conform to the requirements specified in table I.

TABLE I.—*Quantitative requirements*

Characteristics	Requirements		Test reference paragraph
	Minimum	Maximum	
Acidity (percent by weight, as acetic acid)	----	0.01	4.5.1
Distillation range:			
Initial boiling point °C.	110.0	----	
Dry point °C.	----	119.0	4.5.1
Ester content (percent, as isobutyl acetate)	90	----	4.5.2
Nonvolatile matter (g. per 100 ml.)	----	0.002	4.5.3
Specific gravity at 20°/20°C.	0.864	0.873	4.5.1
Color (platinum cobalt scale)	----	10	4.5.4

3.2 Qualitative requirements.

3.2.1 *Appearance.*—The isobutyl acetate shall be clear and free from sediment and suspended matter when examined by transmitted light (see 4.5.1).

3.2.2 *Dryness.*—The isobutyl acetate shall be miscible without turbidity with 19 volumes of 10-degree heptane at 20° C (see 4.4.5).

3.2.3 *Odor.*—The material shall leave the characteristic odor of isobutyl acetate and leave no residual odor after drying on filter paper for 2 hours (see 4.5.1).

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling. — Sampling shall be performed in accordance with method 1021 of Federal Standard No. 141.

4.2.1 *For Military agencies.*—Sampling for Military agencies shall be performed in accordance with method 1031 of Federal Standard No. 141.

4.3 Inspection.—Inspection shall be performed in accordance with method 1021 of Federal Standard No. 141.

4.3.1 *For Military agencies.* — Inspection shall be performed in accordance with method 1031 of Federal Standard No. 141.

4.4 Resubmitted inspection lots. — Paragraph 12.2 of Military Standard MIL–STD–105 shall apply except that a resubmitted inspection lot shall be inspected by the contractor under supervision of the Government inspector using tightened inspection. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be chosen. Before resubmitting, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the inspection lot shall be furnished by the contractor to the Government Inspector.

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4.5 Tests.—Tests shall be made in accordance with the applicable methods of Federal Standard No. 141, and as specified herein.

4.5.1 The following tests shall be conducted in accordance with Federal Test Method Standard No. 141:

TABLE II.

Tests	Method No.
Acidity (see 3.1)	5252
Distillation Range (see 3.1)	4301
Specific Gravity (any convenient method enumerated) (see 3.1)	4183
Appearance (see 3.2.1)	4261
Odor (see 3.2.3)	4401

4.5.2 *Ester content.*—One (1) to two (2) grams of sample shall be placed in a tared ampoule, sealed and weighed to the nearest milligram. The ampoule shall be placed in a 200-millimeter Erlenmeyer flash, containing one and one-half times the theoretical quantity of 0.5*N* alcoholic potassium hydroxide required for complete saponification. The ampoule shall be broken with a stirring rod and a reflux condenser shall be attached to the flask. The flask shall be heated on a steam bath until saponification is complete (1 to 2 hours). During the heating, swirl the flask frequently. Cool, wash down the condenser with distilled water, remove the condenser, and add three drops of phenolphthalein indicator to the contents of the flask. Contents of the flask shall be titrated with 0.5*N* hydrochloric acid. Two blanks of the alcoholic potassium hydroxide solution shall be run along with the sample. The blank shall check within one-tenth of a milliliter. The percentage of ester by weight shall be calculated as follows allowing for acidity as determined in 4.5.1 (see 3.1).

$$\frac{((\text{ml. KOH} \times \text{N.}) - (\text{ml. HCl} \times \text{N.})) \times 11.6}{\text{WT of sample}} = \text{Percent ester as isobutyl acetate}$$

4.5.3 *Nonvolatile matter.*—Place 100 milliliters of the sample in a tared evaporating dish and evaporate over a steam bath almost to dryness. Heat the dish and residue in an oven at 105° ± 2° C. (221° ± 4° F.) to constant weight, cool in a desiccator and weigh. Report the difference in weight as grams of nonvolatile matter per 100 milliliters of the sample (see 3.1).

4.5.4 *Color.*—The color of the isobutyl acetate shall be determined in accordance with Method D1209–54 of the American Society for Testing Materials Standards (see 3.1).

4.5.5 *Dryness.*—Transfer 5 milliliters of the sample to a 100-milliliter glass-stoppered cylinder and add normal heptane in 5 milliliter portions until the total volume specified is obtained. Both the sample and the normal heptane shall be kept at 20° C. (68° F.) during the test and the mixture shall be shaken well after each addition. Turbidity of the mixture indicates the presence of excessive water in the sample (see 3.2.2).

5. PREPARATION FOR DELIVERY

For civil agency procurements, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level C.*—Isobutyl acetate in quantity specified shall be packaged in accordance with the manufacturers' commercial practice.

5.2 Packing.

5.2.1 *Level C.*—Isobutyl acetate packaged as specified shall be packed in containers which will insure acceptance by common carrier and safe delivery to destination at the lowest applicable rate. Shipping containers shall conform to the Uniform Freight Classification Rating Rules and Regulations or other regulations as applicable to the mode of transportation.

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5.3 **Marking.**

5.3.1 *Civil agencies.* — Marking for civil agencies shall be in accordance with Federal Standard No. 123.

5.2.3 *Military agencies.* — In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.**—This specification covers isobutyl acetate, an ingredient for use in specifications for paints and related materials and as such is not intended for use for direct procurement by the Government. In the event of small purchases for experimental purposes and for purchases by industry, suitable preparation for delivery requirements should be made a part of the contract. The quality assurance provisions are provided for use in connection with inspection of material when used as an ingredient in paints purchased for the Government.

6.2 **Basis of purchase.**—The material shall be purchased by volume, the unit being a U. S. gallon (231 cubic at 15.5° C. (60° F.)).

6.3 **Ordering data.** — Procurement documents should state the title, number, and date of this specification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C
Navy—A Sh Or MC Md
Air Force.

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